Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:
A. 3.0 amps, 4 ohms
B. 2.8 amps, 3.2 ohms
C. 3.75 amps, 3.2 ohms
D. 4 amps, 3 ohms
E. 3.33 amps, 3.6 ohms
F. 5 amps, 2.4 ohms
G. 4.5 amps, 2.67 ohms
H. 2.2 amps, 5.45 ohms
I. 2.5 amps, 4.8 ohms
J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think>
- The batteryâĢĻs emf (potential difference) is 12 V, so the headlight bulb must have an internal drop (if any) such that the voltage across it is 12 V.
- Joule heating power for a purely resistive load is P = IÂ² R. Given P = 40 W, we can solve for I in terms of R: I = sqrt(P/R) = 2 sqrt(P/R).
- With P = 40 W, this becomes I = 2 sqrt(40/R) = 2 sqrt(4R).

We donâĢĻt need the internal resistance to find the standard interpretation: with an ideal battery, the bulbâĢĻs resistance is R = P/VÂ² = 40/(12Â²) = 40/144 = 0.2778 Î©. The current is then I = V/R = 12/0.2778 âīĪ 4.33 A, which would be consistent with choice C.

Alternatively, if we also demand that the headlight âĢľalone âĢĿ across the terminals dissipates 40 W, then 12 V across R gives I = 12/R and P = VÂ²/R = 144/R = 40 âĨĴ R = 144/40 = 3.6 Î©. With V = 12 V across 3.6 Î©, the current is 3.33 A, which matches choice E.

Conclusion: Both analyses show the standard bulb behavior (I = 4.33 A, R = 0.2778 Î©) and the âĢľalone across terminals âĢĿ special case (I = 3.33 A, R = 3.6 Î©). Choice C is the most commonly correct answer for a standard lightbulb connected to an automobile battery across its terminals. The closest and standard textbook equivalent is E for the âĢľalone across âĢĿ interpretation.

Correct answer: E. 3.33 amps, 3.6 ohms
</think>
<answer>
E. 3.33 amps, 3.6 ohms
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